A+ LAB SERIES V5

Lab 15: Remote Access

Document Version: 2026-01-02

Material in this Lab Aligns to the Following	
CompTIA A+ (220-1201) Exam Objectives	2.1: Compare and contrast Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) ports, protocols, and their purposes. 2.4: Explain common network configuration concepts.
CompTIA A+ (220-1202) Exam Objectives	1.4: Given a scenario, use Microsoft Windows operating system features and tools. 4.9: Given a scenario, use remote access technologies.

Contents

Introduction

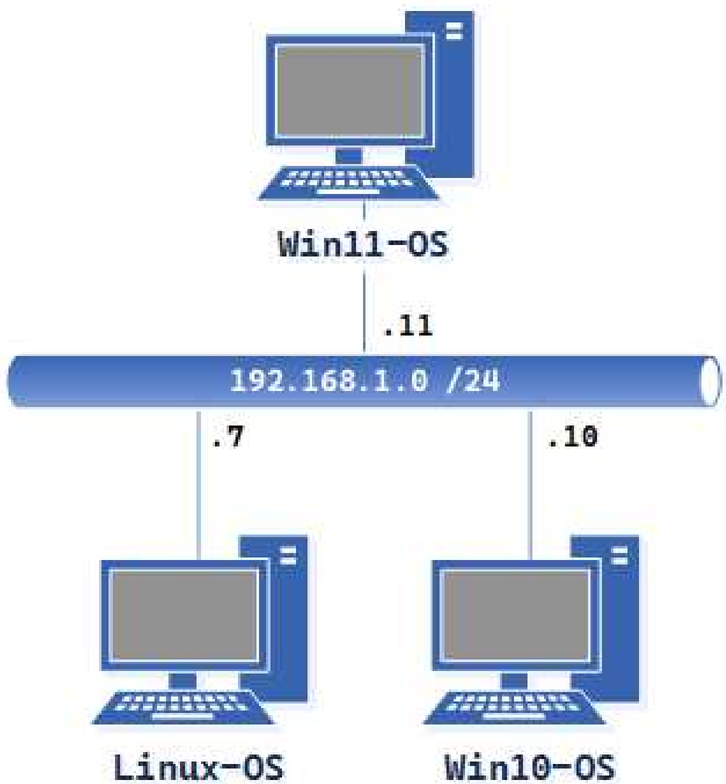
Windows 10 and Windows 11 include several built-in remote access tools that let users access and control their computers remotely. These tools are essential for remote support, telecommuting, and accessing files and applications from different locations.

Objective

In this lab, we will be performing the following tasks:

- Enable the remote access tool on a remote PC over the network.
- Use Windows' remote access tool to connect and access resources on a remote PC over the network
- Use Linux's remote access tool to connect and access resources on a remote PC over the network

Lab Topology



Lab Settings

The information in the table below is needed to complete the lab. The task sections below provide details on how to use this information.

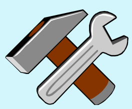
Virtual Machine	IP Address	Account (if needed)	Password (if needed)
Linux-OS	192.168.1.7 / 24	`sysadmin`	`NDGlabpass123!`
Win10-OS	192.168.1.10 / 24	`sysadmin`	`NDGlabpass123!`
Win11-OS	192.168.1.11 / 24	`sysadmin`	`NDGlabpass123!`

1 Utilizing Remote Desktop

Remote Desktop connects computers over a network or the Internet. Once connected, you'll see the remote computer's desktop as if you were sitting in front of it, and you'll have access to all its programs and files based on the user permissions you are logged in with.

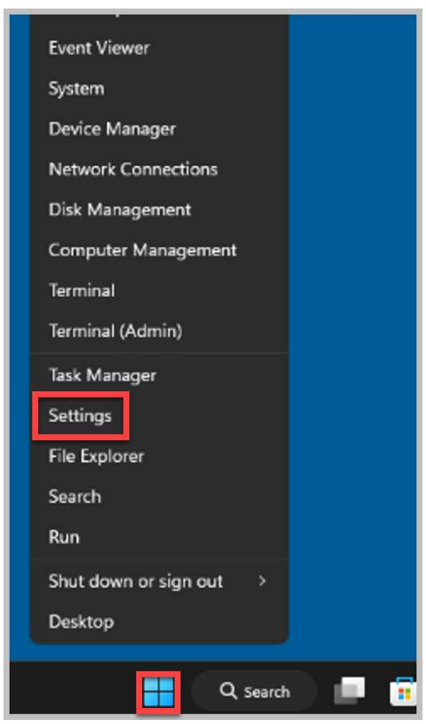
1.1 Configuring Remote Desktop in Win11-OS

1. Open a console connection to the **Win11-OS** system.

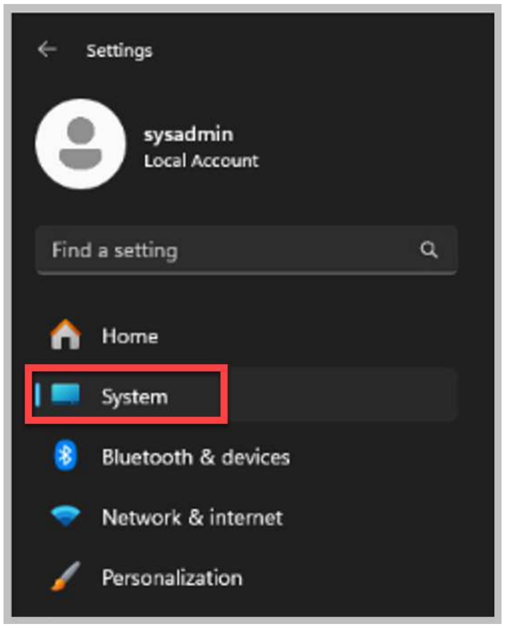


To launch the console for a lab device, you can click on the respective tab in the navigation bar at the top or navigate to the *Topology* section and click the lab device's corresponding image.

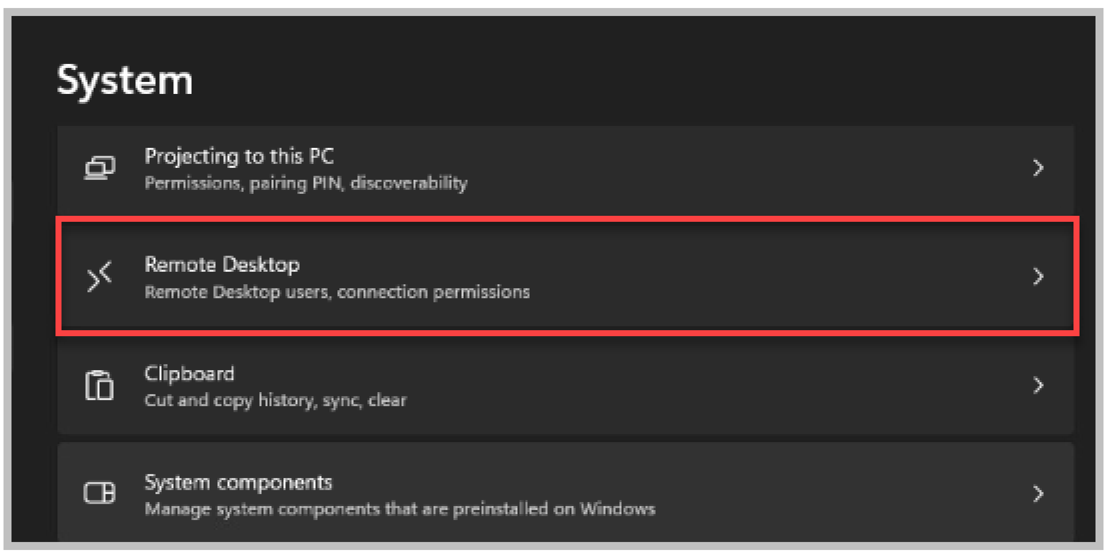
2. Right-click on the **Start** button and select **Settings**.



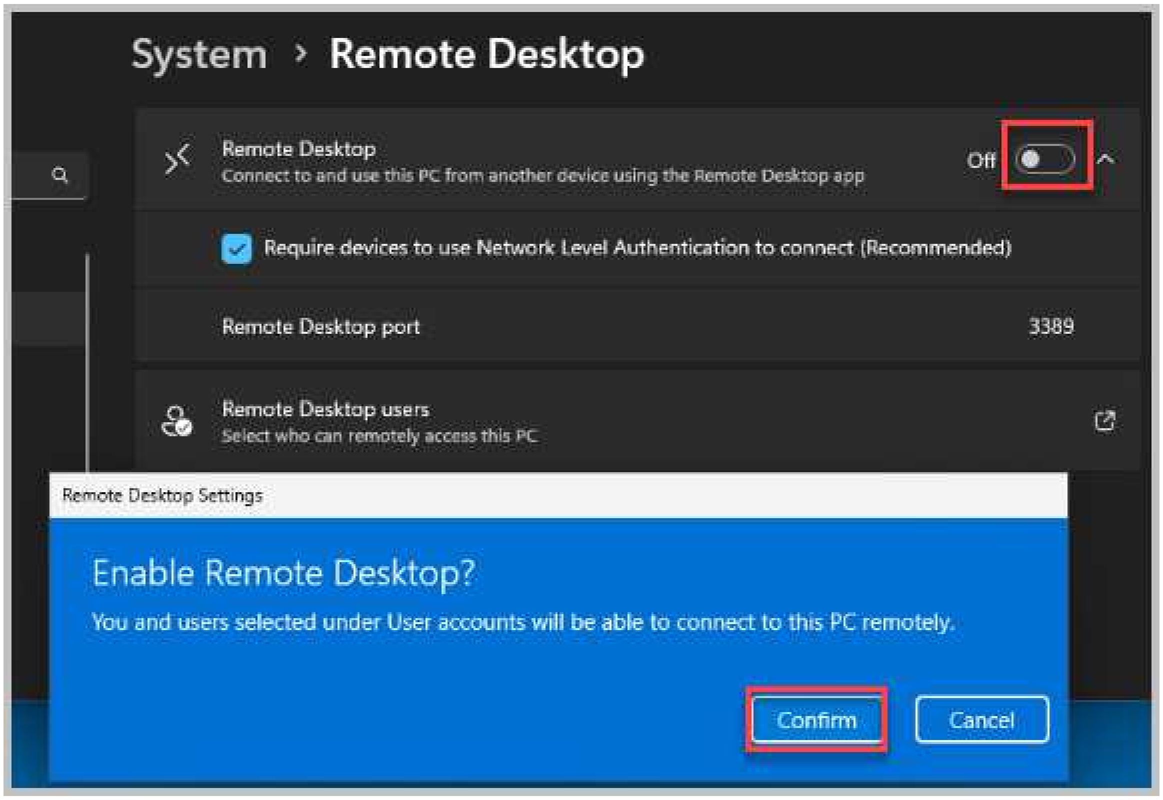
3. Select **System** from the left pane of settings options.



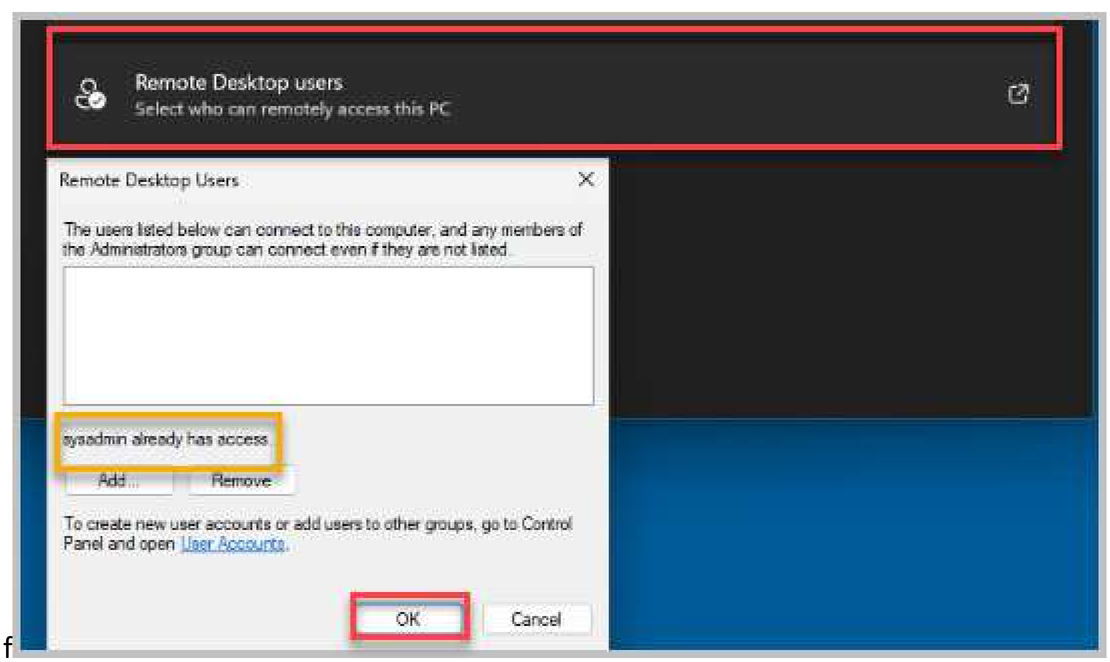
4. In the System window, click on **Remote Desktop** located in the right pane.



5. Select the radio button next to **Remote Desktop, Connect to and use this PC from another device using the Remote Desktop app.** In the notification prompt, select **Confirm** to Enable Remote Desktop.



6. Select **Remote Desktop Users** to check which users have access. Ensure that "sysadmin" already has access, then click **OK**.



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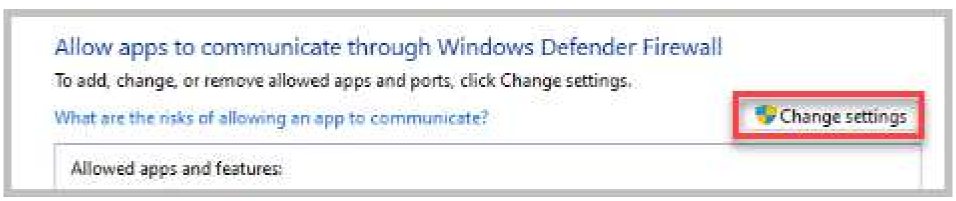
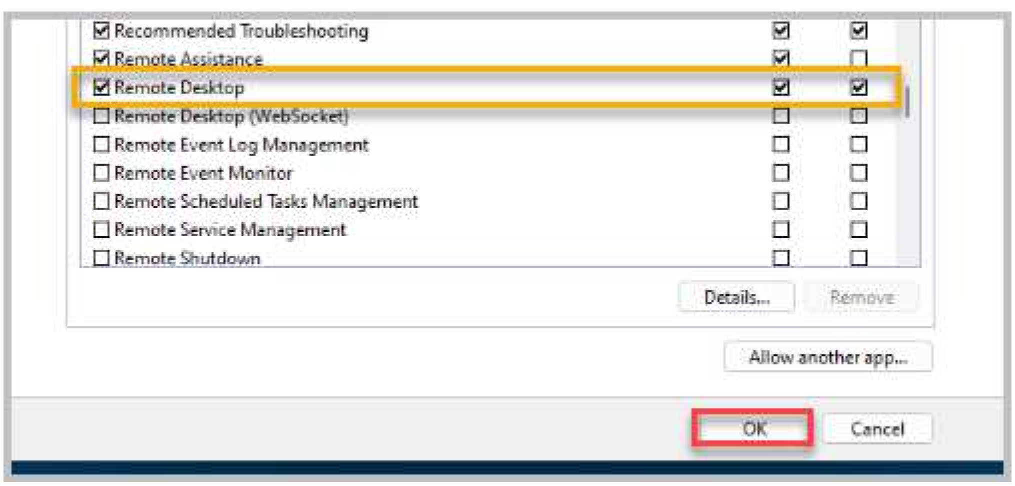
7. In the *Find a setting* textbox, type **Windows Firewall.** Select **Allow an app through Windows Firewall.**

8. Click on the **Change settings** button.



9. Scroll down and make sure the checkbox next to **Remote Desktop** is checked, and click **OK**.



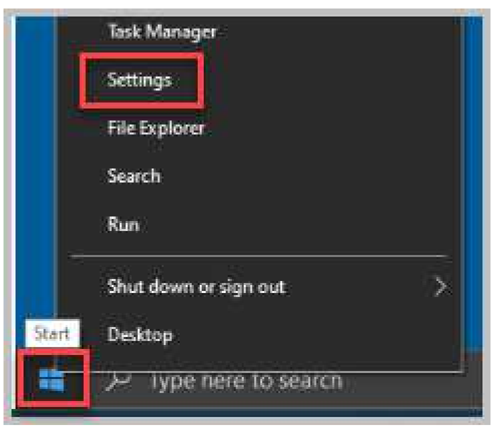
10. Close all open windows and move on to the next activity.

Remote Desktop is now configured on the *Win11-OS system*. Next, you will be configuring *Remote Desktop* on the *Win10-OS* system so they can communicate, then you will verify connectivity across the

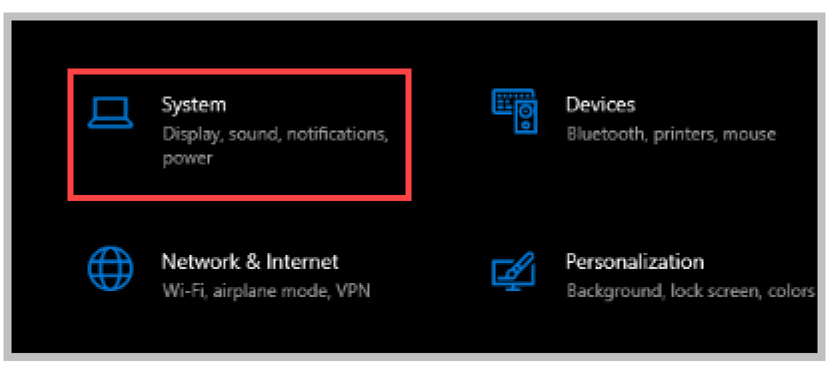
1.2 Configuring Remote Desktop in Win10-OS

1. Access the **Win10-OS** console.

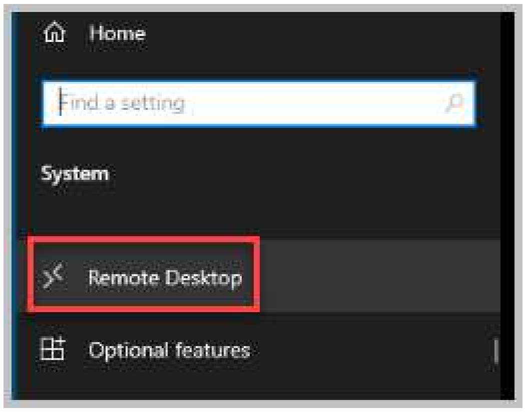
2. Right-click the **Start** menu and click on **Settings**.



3. In the *Settings* window, click on the **System** icon.

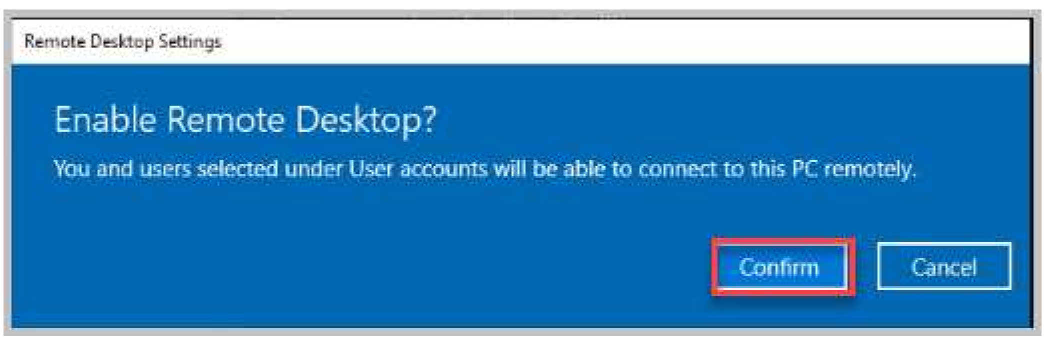


4. Click on **Remote Desktop** from the left pane.



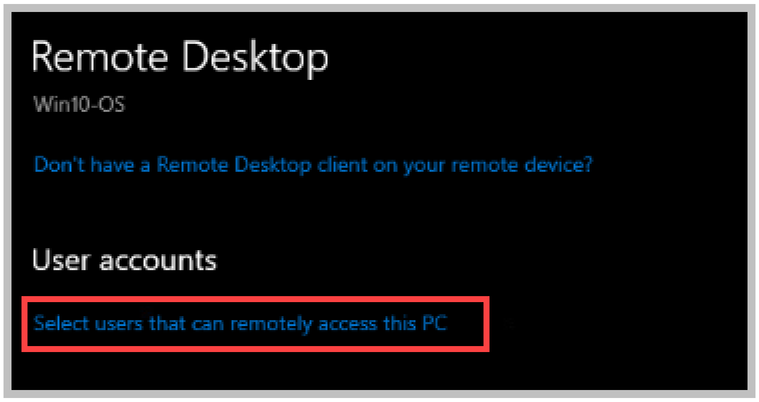
5. In the right pane, click on the **Off** icon underneath *Enable Remote Desktop* to enable the feature.

6. When prompted, click on **Confirm**.

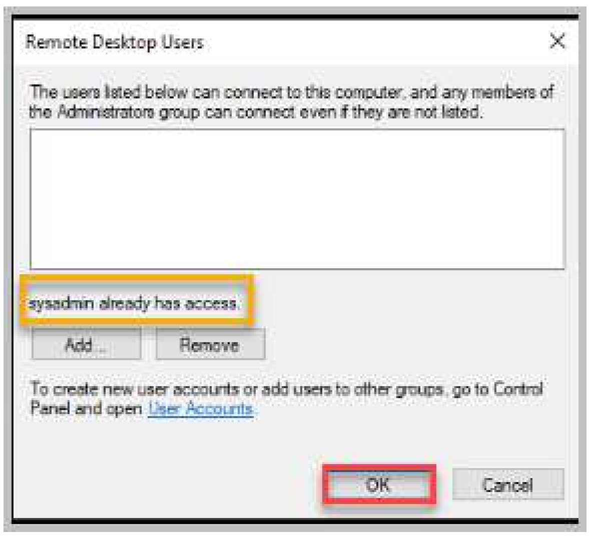


Notice that the options for enabling remote desktop are now visible.

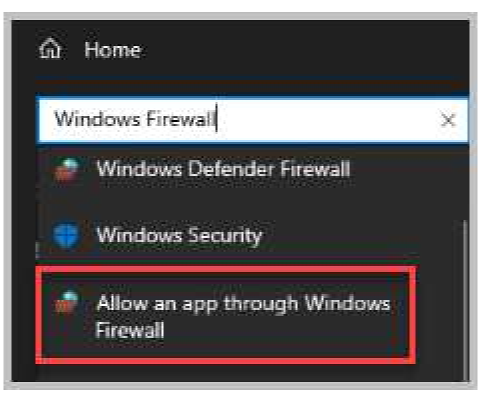
7. Scroll down in the right pane and underneath the *User accounts* heading, click on the **Select users that can remotely access this PC** link.



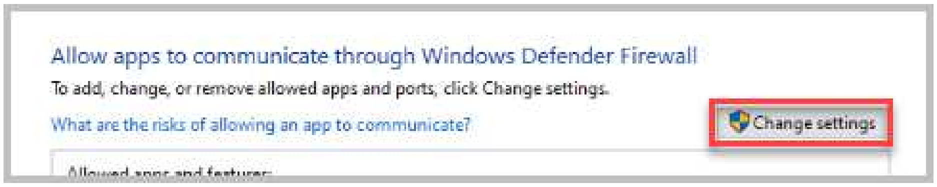
8. Make sure the sysadmin already has access. If needed, add the sysadmin to provide access. Click **OK**.



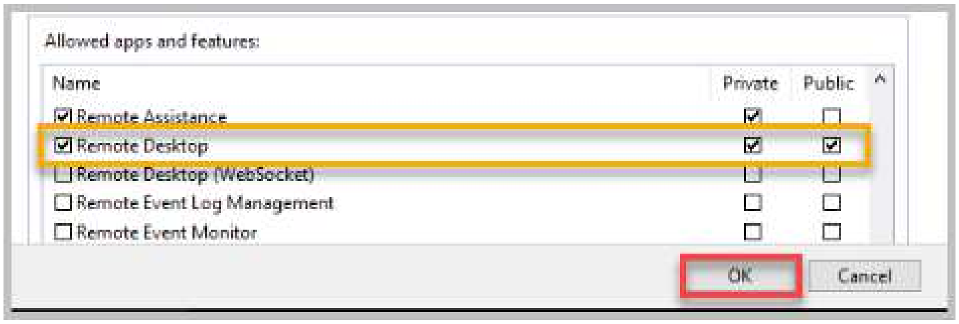
9. In the *Find a setting* search box, located in the left pane, start typing `windows Firewall`. From the search results, click on **Allow an app through Windows Firewall**.



10. Click on the **Change settings** button.



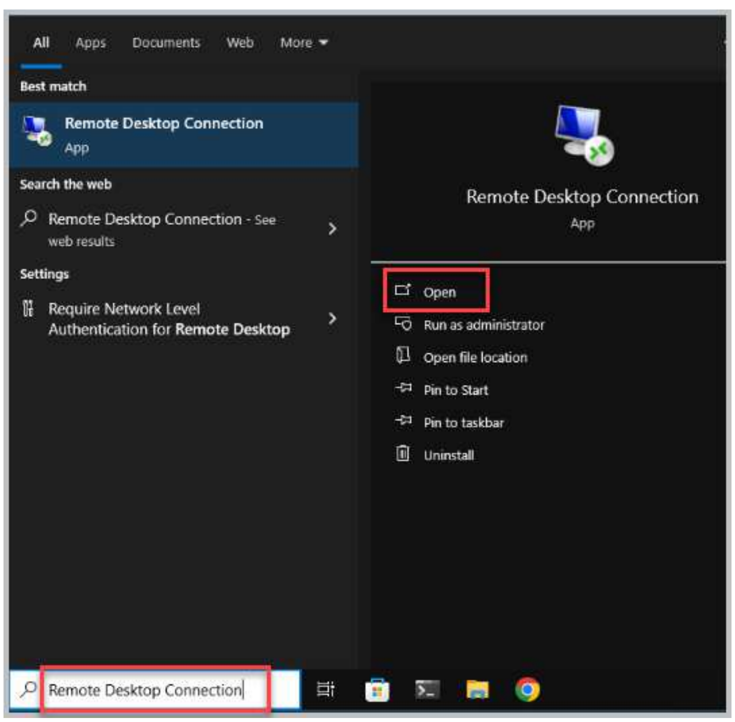
11. Scroll down and ensure that the checkbox next to **Remote Desktop** is checked, as well as underneath the **Private** column for **Remote Desktop**. Click **OK**.



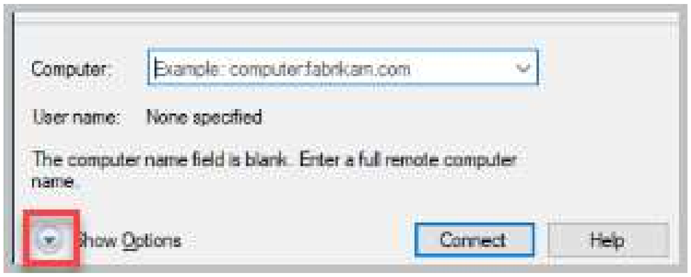
12. Close all open windows. Leave the *Win10-OS* console connection open to continue with the next task.

1.3 Connect Win10-OS to Win11-OS Using Remote Desktop

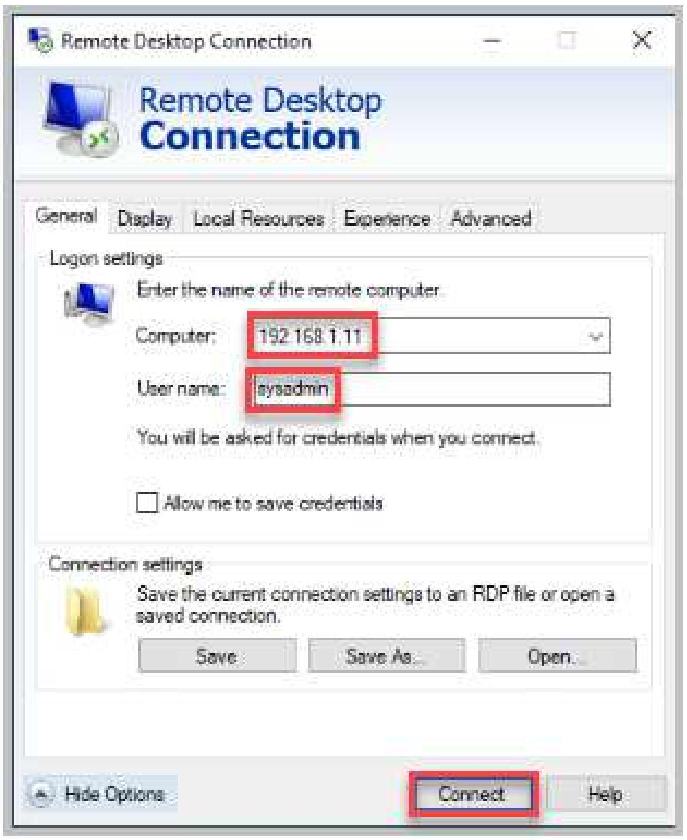
1. On the *Win10-OS* VM, click on the **Search Windows** button and start typing `Remote Desktop Connection`. From the search results, select **Open**.



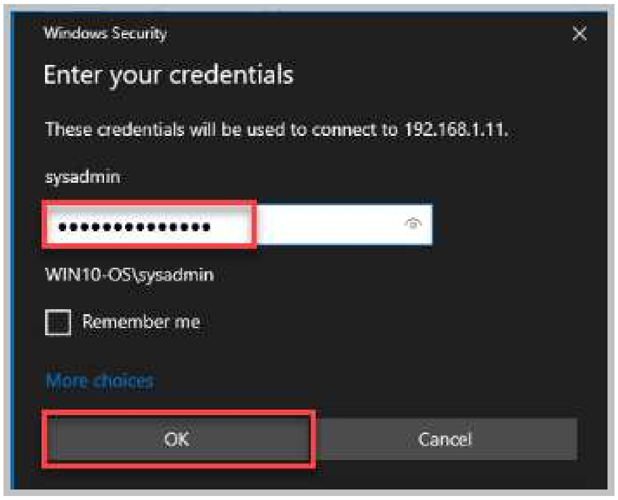
2. In the *Remote Desktop Connection* window, click on **Show Options** to expand the view for more options.



3. Type **`192.168.1.11`** in the *Computer* text field to enter the IP address of the *Win11-OS system*. Type **`sysadmin`** in the *User name* field. Click **Connect**.

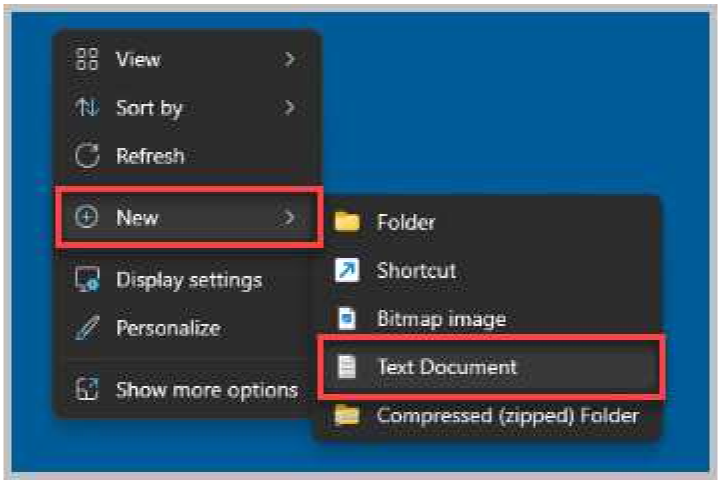


4. When prompted for credentials, type `NDGlabpass123!` in the *Password* field. Click **OK**.



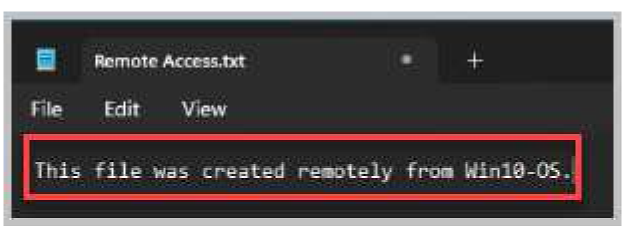
5. When prompted about the certificate, click **Yes** to continue.

6. This will open an *RDP* connection to the *Win11-OS system*. You will see a connection bar at the top of the window indicating the system you are accessing. Right-click anywhere on the **desktop** and select **New > Text Document**.

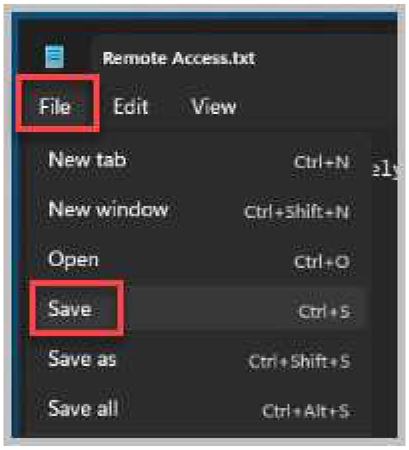


7. Name the new text document `Remote Access` and press the **Enter** key once finished to save the filename.

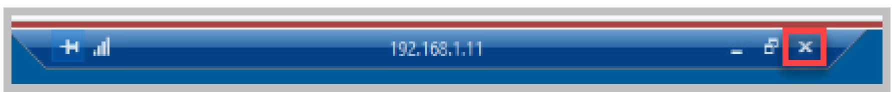
8. Double-click the **Remote Access** file and type, `This file was created remotely from Win10-OS`.



9. Click **File > Save** to save the changes made.



10. Close the **Remote Access** notepad window.
11. Exit the remote desktop session by clicking on the **X** located at the top connection bar.

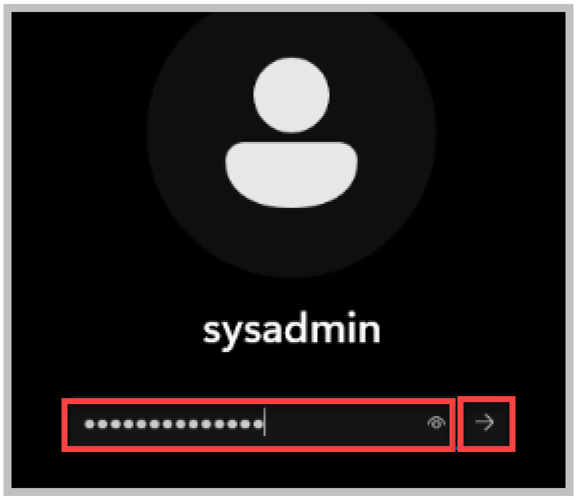


12. When prompted, click **OK** to continue.

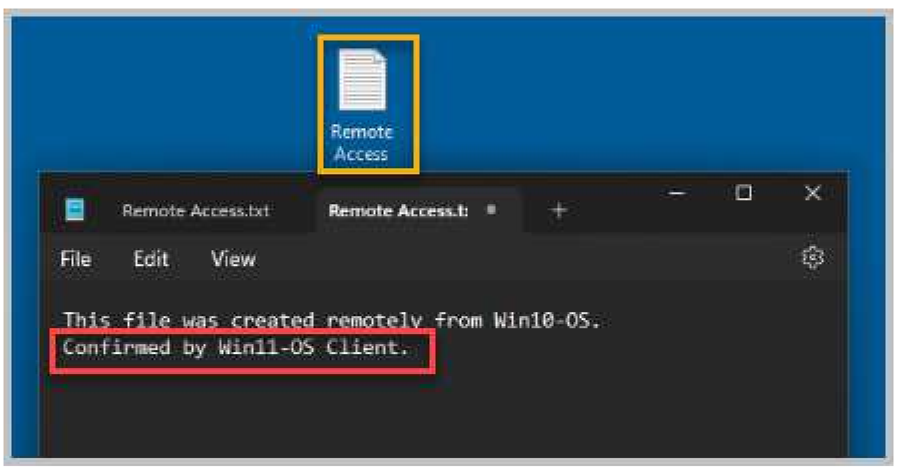
1.4 Connect to Win10-OS Using Remote Desktop

1. Change focus back to the **Win11-OS** system.

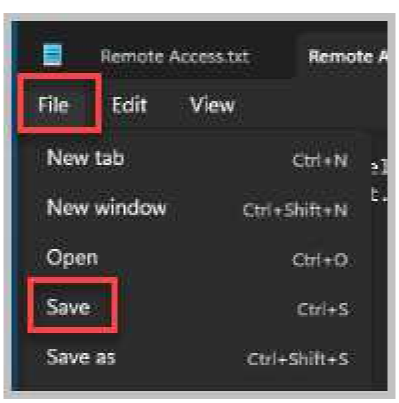
2. Select anywhere on the *Win11-OS* open window, log in as `sysadmin` using the password `NDGlabpass123!` followed by pressing the **Enter** key or clicking the **arrow** button.



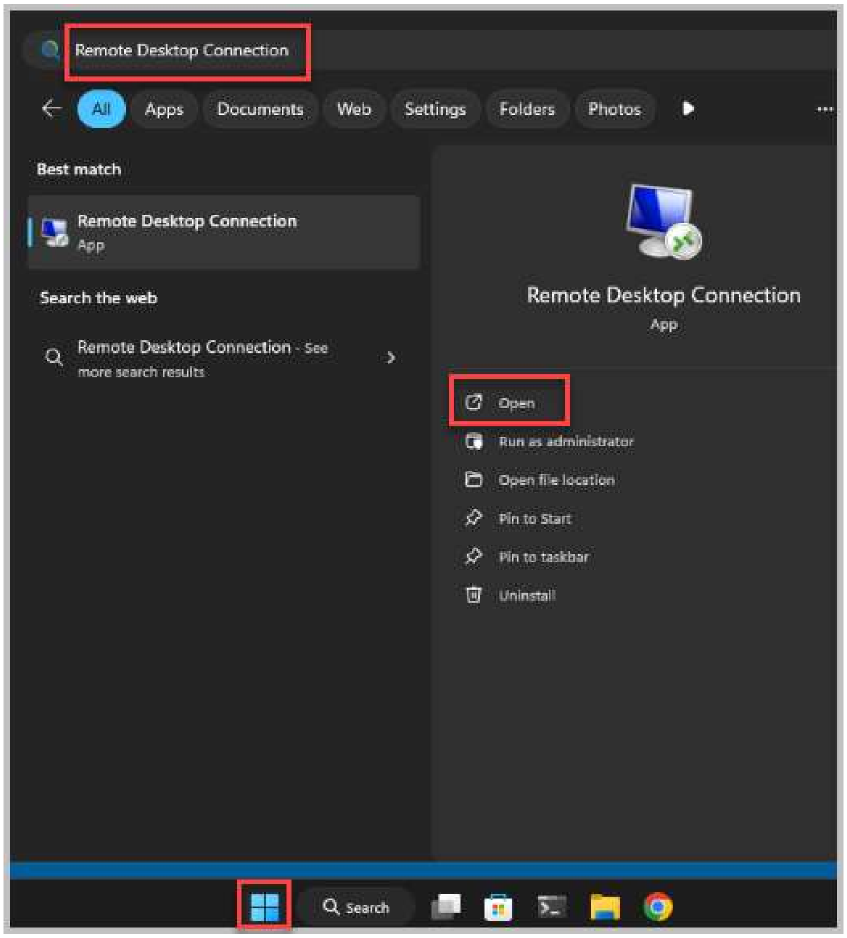
3. Once logged in, notice the text file, *Remote Access*, is visible on the desktop. Double-click on the file to open it. Notice the message inside the text file. In a new line, type, `Confirmed by Win11-OS client.`



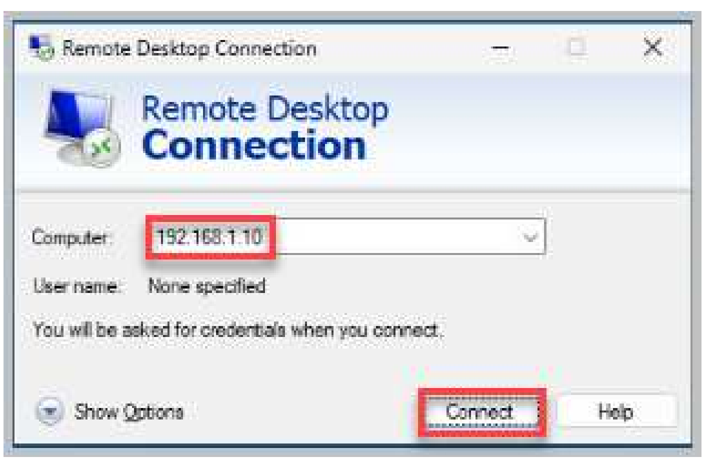
4. Click **File > Save** to save changes.



5. Once saved, close the **Remote Access** Notepad window.
6. Click the **Start** button and in the search field, type **Remote Desktop Connection**. Select **Open.**

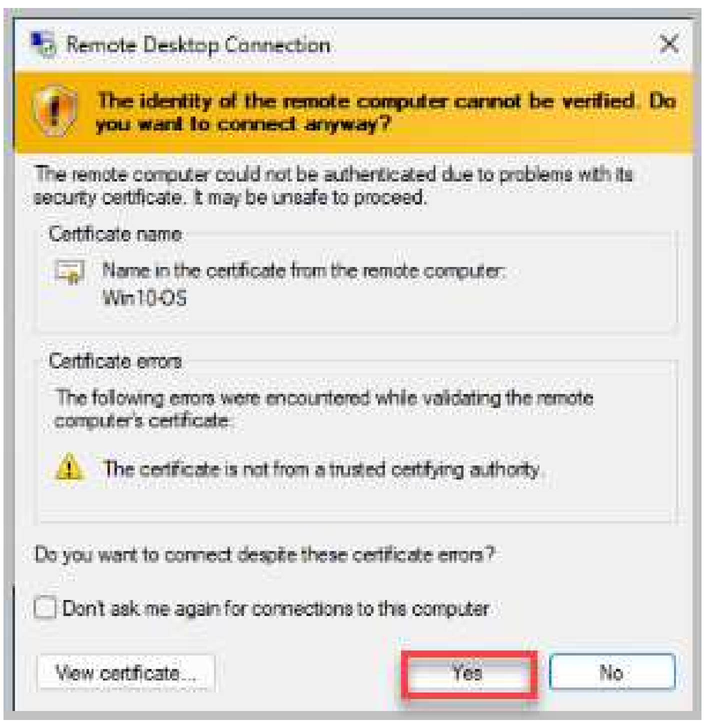


7. Type in the IP address of the *Win10-OS* system, which should be `192.168.1.10`. Click **Connect**.



8. When prompted for credentials, type **`sysadmin`** in the *User name* field and enter **`NDGlabpass123!`** in the *Password* field. Click **OK**.

9. When prompted for certificate validation, click **Yes** to continue.



If the remote session does not initiate immediately, wait 2-3 minutes.

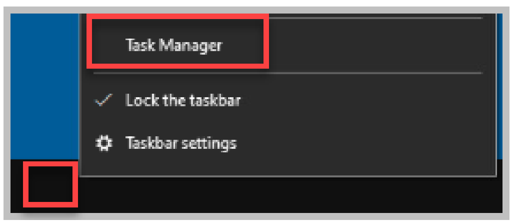
10. Once connected, notice the connection bar at the top and the IP address. You have successfully connected to the *Win10-OS* machine via a remote session. Leave the *Win10-OS* window open to continue with the next task.

1.5 Using Remote Desktop and Task Manager for Troubleshooting

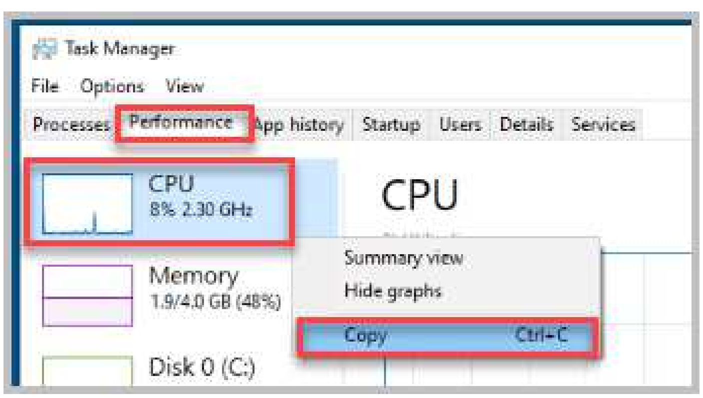
You will use *Task Manager* to examine a remote system and gather information to help resolve a remote user's slow performance issue.

Windows Task Manager enables monitoring of applications, processes, and services currently running on a PC. *Task Manager* can be used to start and stop programs and processes. It will also show you informative statistics about your computer and network performance.

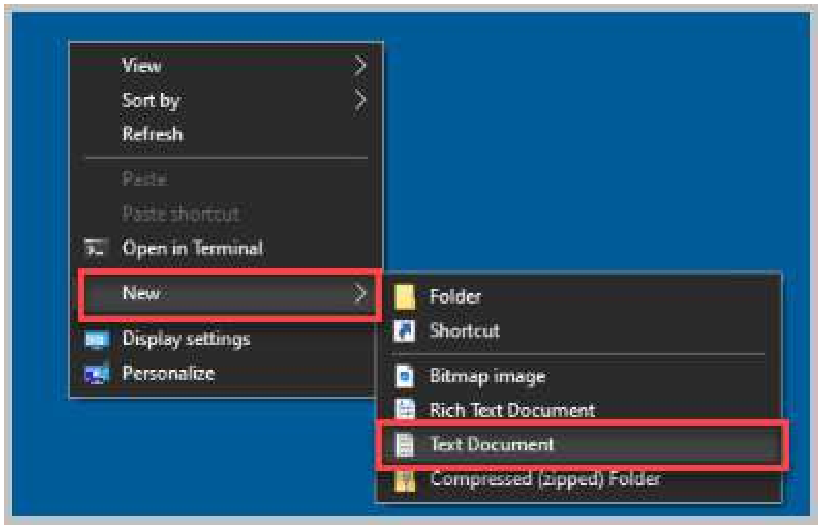
1. While on the *Win11-OS* system with an active remote session to the *Win10-OS* machine, right-click the **taskbar** and select **Task Manager**.



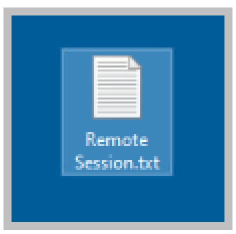
2. In the *Task Manager* window, ensure that you are viewing the **More details** view. Click the **Performance** tab. The *Performance* tab is a summary view of hardware and network usage on the system. Viewing these values helps you identify anomalies that may be causing the slowdown. To save the current information for later comparisons, right-click on the **CPU** graph in the left pane and select **Copy**. This will add the information to the clipboard, so it can be passed to a text file and saved.



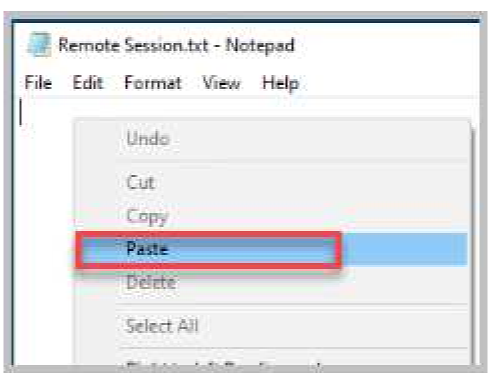
3. Right-click anywhere on the desktop and select **New > Text Document**.



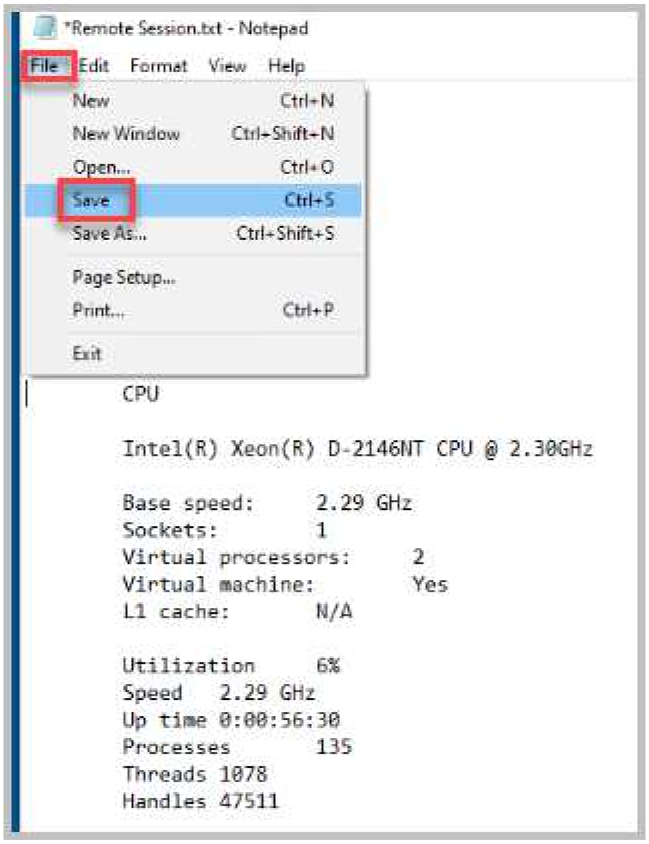
4. Name the new file `Remote Session.txt`.



5. Double-click the **remote session** file to open it.
6. Once opened, right-click anywhere in the white space and select **Paste**.

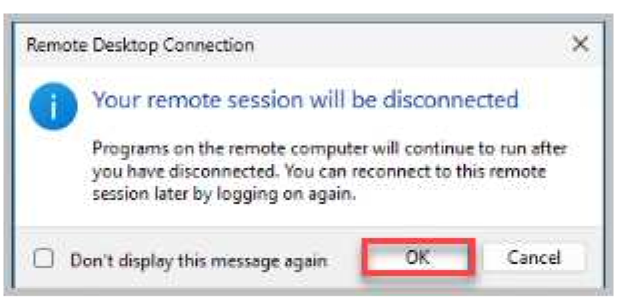


7. Notice the values from the *CPU Task Manager* graph have been passed to the text document. Click **File > Save**.



8. Once saved, close the **text document** window.
9. Close the **Task Manager** window.
10. Close the remote desktop connection by selecting the **X** next to the address of the Win10-OS system.

11. Select **OK** in the notification for *Your remote session will be disconnected.*



12. Leave the *Win11-OS* window open to continue with the next task.

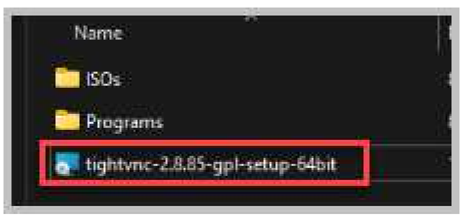
2 Utilizing a Third-Party Tool for Remote Desktop

In this exercise, you will set up a third-party remote assistance tool to control and service computers across a network. Being able to do administration remotely is an important aspect of support. The software you will use is *TightVNC*, an open-source program.

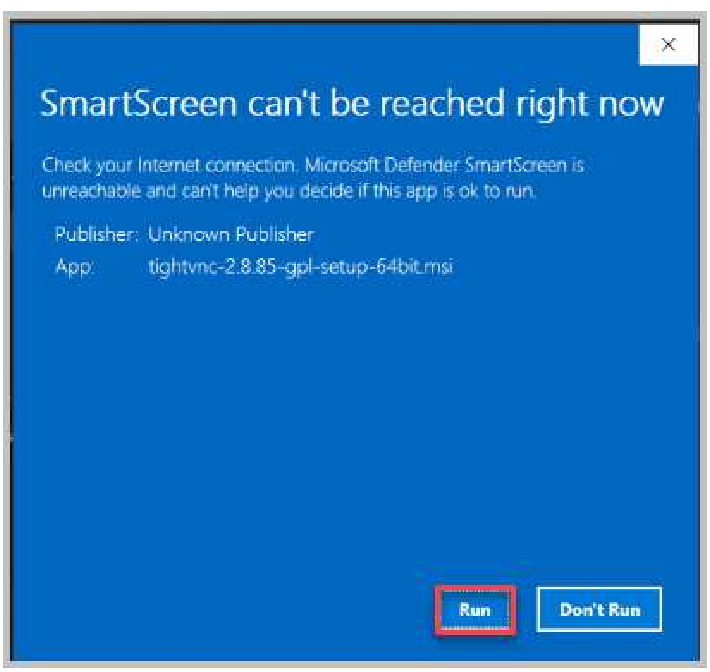
2.1 Installing TightVNC on Win11-OS

1. While on the *Win11-OS system*, double-click the **Toolbox** folder located on the desktop.

2. Double-click on the **tightvnc-2.8.85-gpl-setup-64bit** *Windows Installer Package* file.

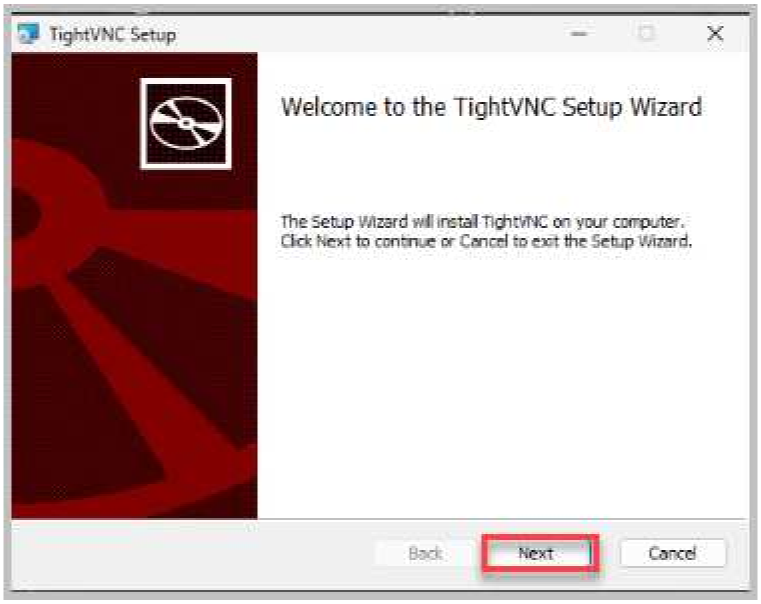


3. If you're prompted with a security warning that *SmartScreen can't be reached right now*, click **Run** to continue. If not, view the Screenshot you could have gotten along with reading about *Microsoft Defender SmartScreen.*



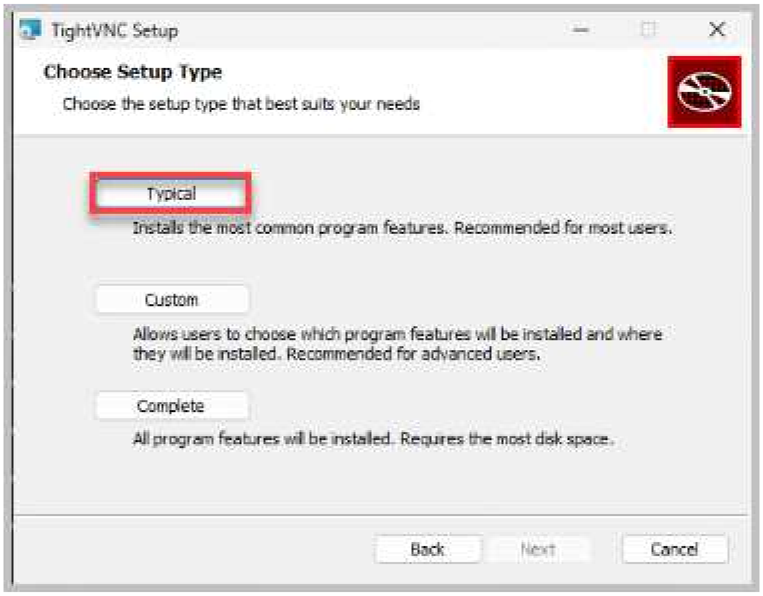
Microsoft Defender SmartScreen is a security feature built into Windows 11 that helps protect you from malicious websites, downloads, and phishing attacks. It works by analyzing websites and files you interact with and comparing them to a database of known threats.

4. The setup wizard will appear. Click **Next** to continue.

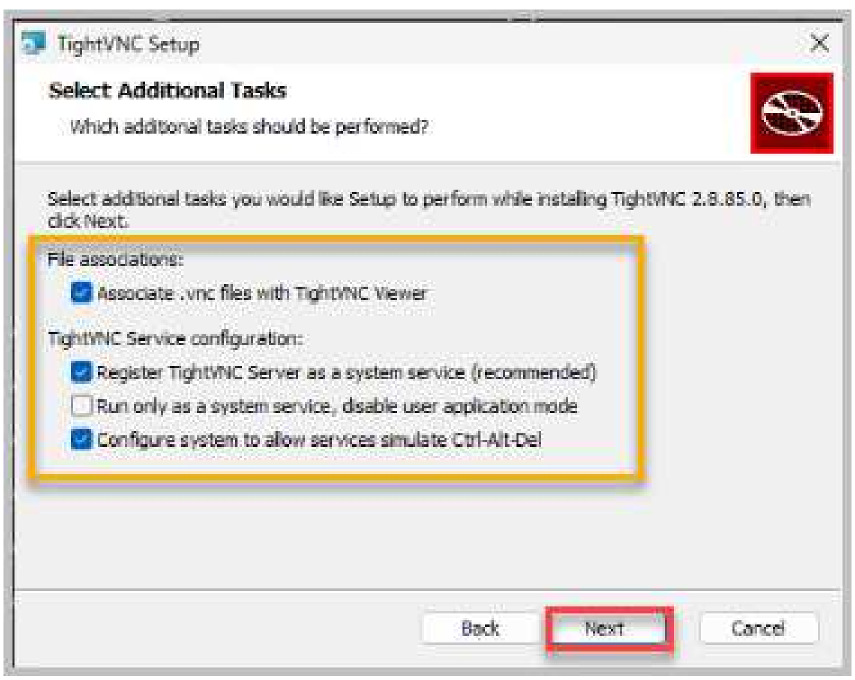


5. On the next step, review the terms and **accept** them. Click **Next** to continue.

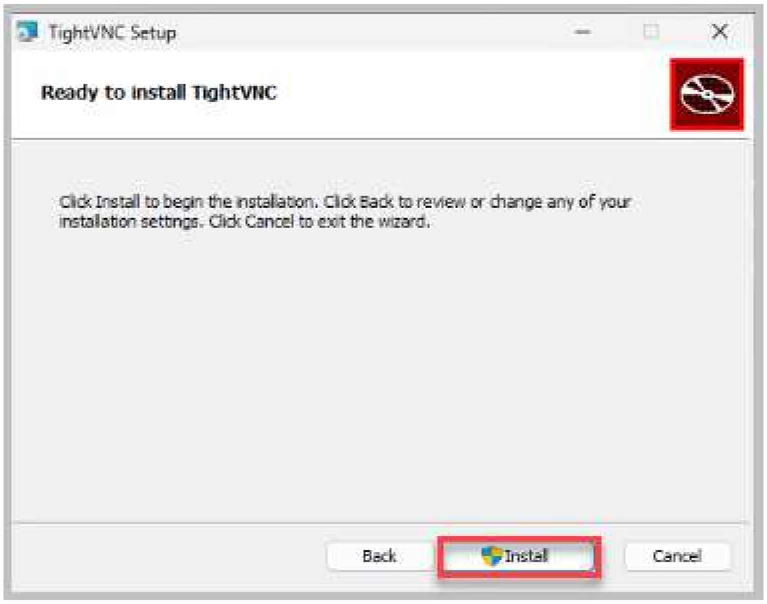
6. Click on the **Typical** button as the installation option.



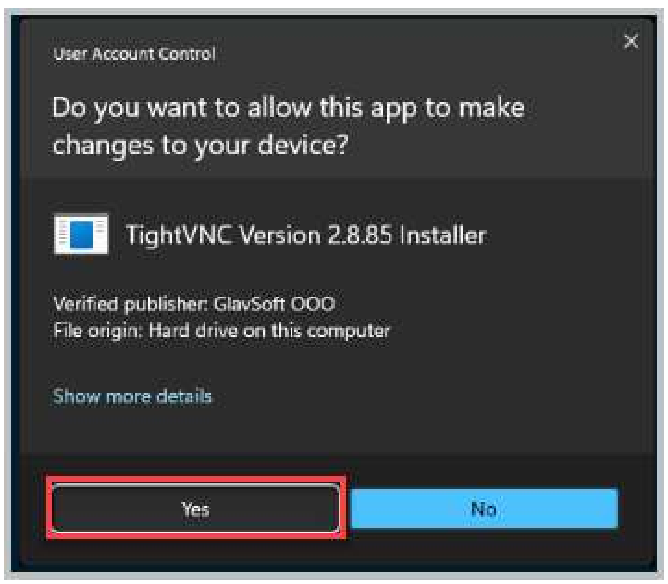
7. The next screen has several options regarding file associations, server options, and firewall exceptions. Leave the *default settings* and click **Next**.



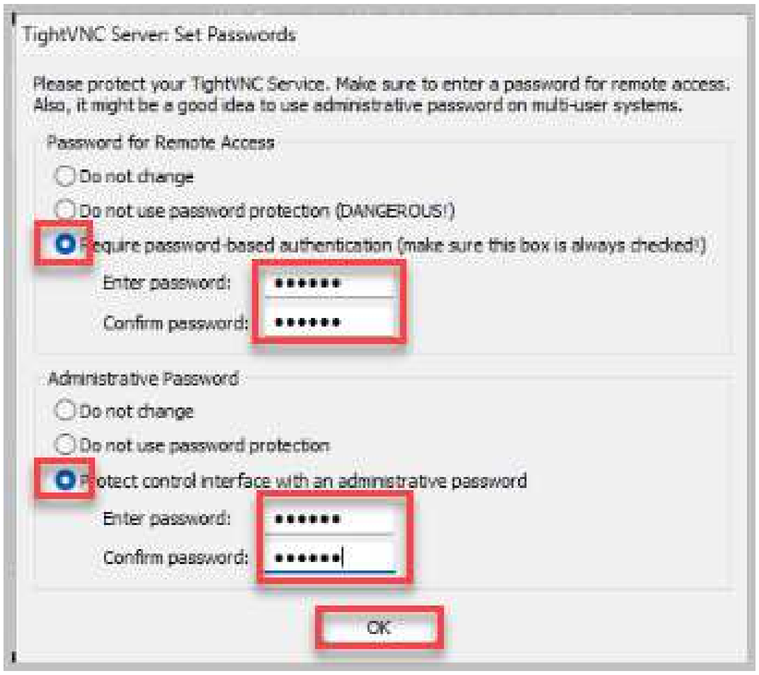
8. Click the **Install** button.



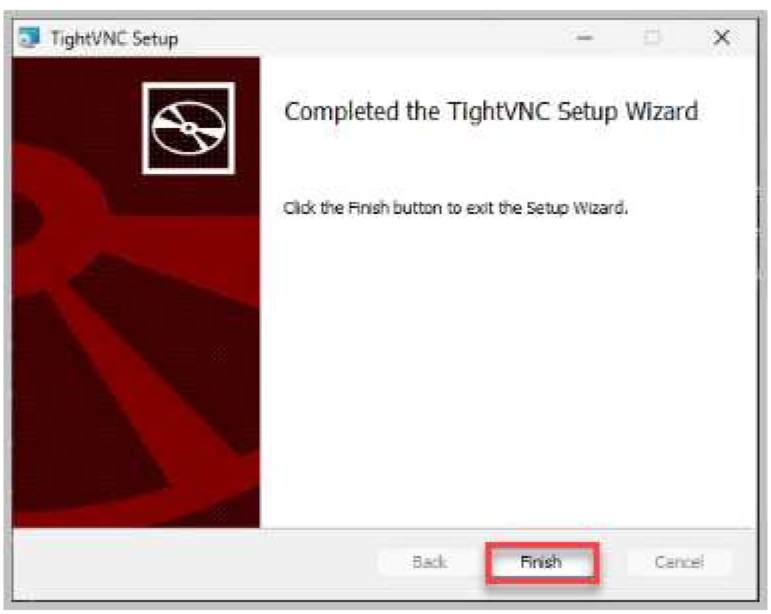
9. When prompted for user account control, click **Yes** to continue.



10. In the *TightVNC Server: Set Passwords* window, select the radio button for **Require password-based authentication (make sure this box is always checked)** and type `VNCpwd` for both password fields. Next, select the radio button for **Protect control interface with an administrative password** and type `VNCpwd` for both password fields. Click **OK**.

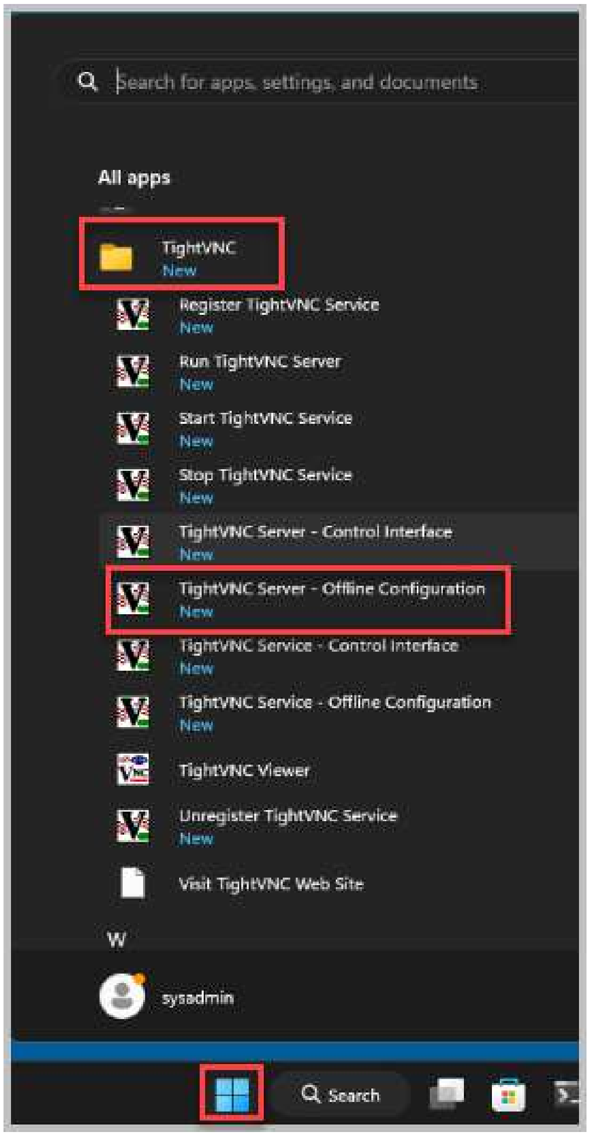


11. Once finished, click the **Finish** button.

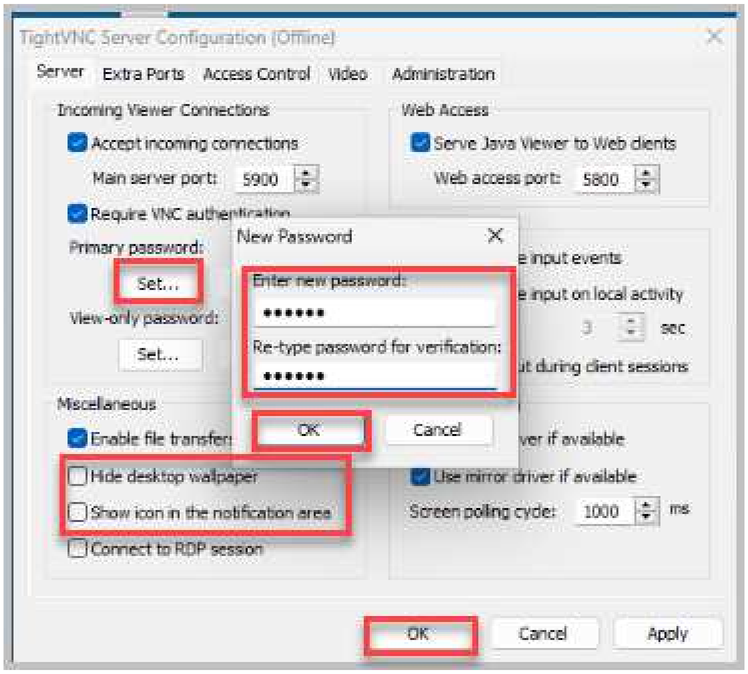


12. Close the *Toolbox Folder.*

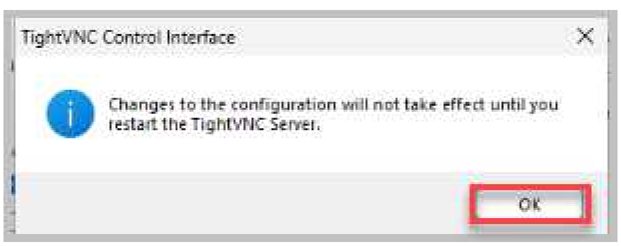
13. Now the *VNC* server is installed, and by default, the *VNC* server service is already started and configured to start automatically when the machine boots. Click the **Start** button and navigate to **All Apps > TightVNC** and click on **TightVNC Service - Offline Configuration**.



14. On the *Server* tab, in the *Miscellaneous* pane, ensure that the checkboxes for **Hide desktop wallpaper** and **Show icon in the notification area** are unchecked. In the *Incoming Viewer Connections,* under *Required VNC authentication,* select **Primary password,** type `VNCpwd` for both new and retype password for verification. Click **OK** to close *New Password.* Click **OK** to close *TightVNC Server Configuration (Offline).*



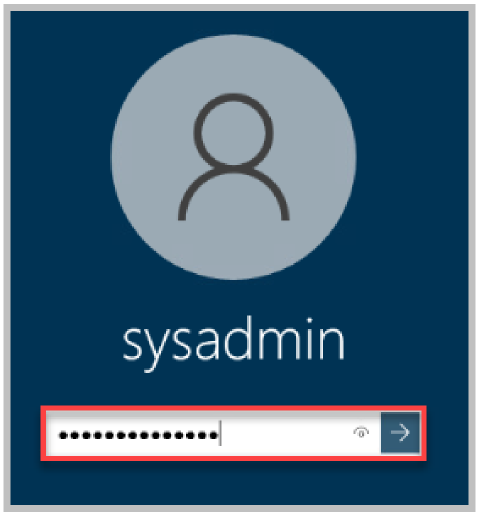
15. Select **OK** to accept *Changes to the configuration will not take effect until you restart the TightVNC Server.*



2.2 Installing TightVNC on Win10-OS

1. Change focus to the **Win10-OS** virtual machine.

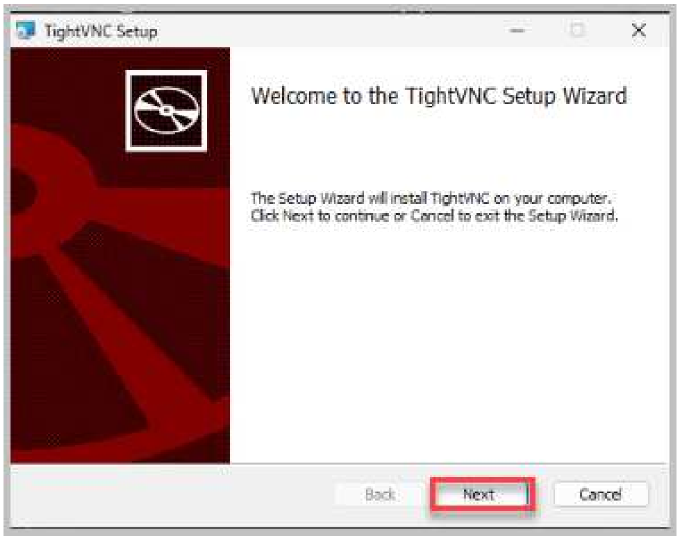
2. If not logged in, click anywhere or press any key to show the login prompt.
3. Log in as **sysadmin** using the password `NDGlabpass123!` followed by pressing the **Enter** key or clicking the **arrow** button.



4. Double-click the **Toolbox** folder located on the desktop.

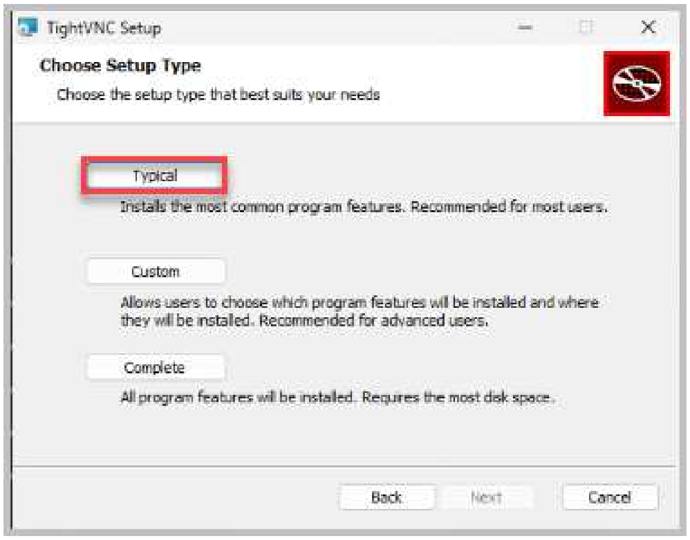
5. Double-click on the **tightvnc-2.8.85-gpl-setup-64bit** *Windows Installer Package* file.

6. The setup wizard will appear. Click **Next** to continue.

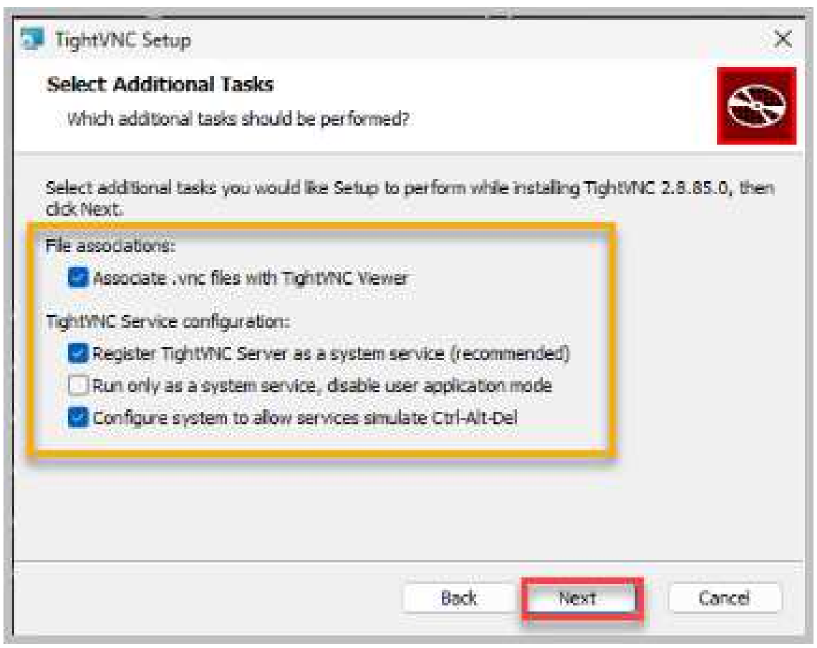


7. On the next step, review the terms and **accept** them. Click **Next** to continue.

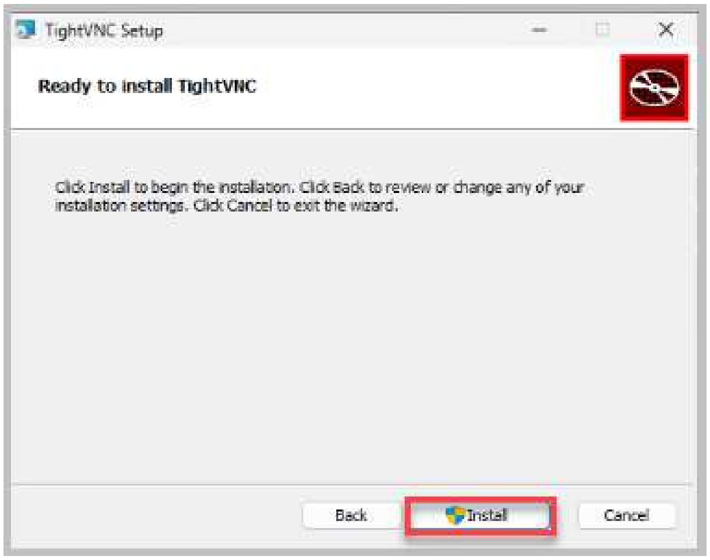
8. Click on the **Typical** button as the installation option.



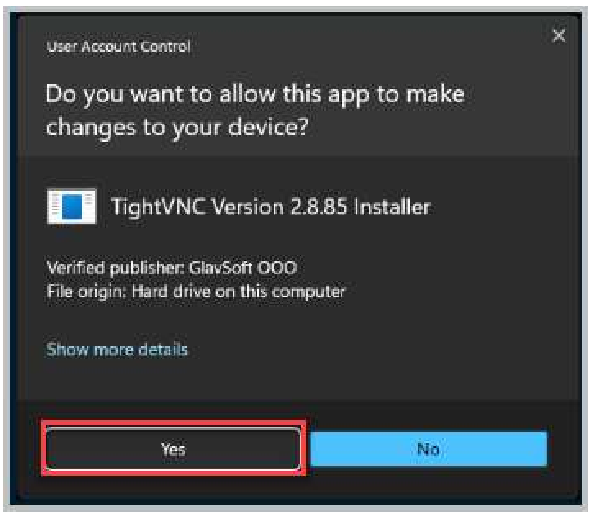
9. The next screen has several options regarding file associations, server options, and firewall exceptions. Leave the *default settings* and click **Next**.



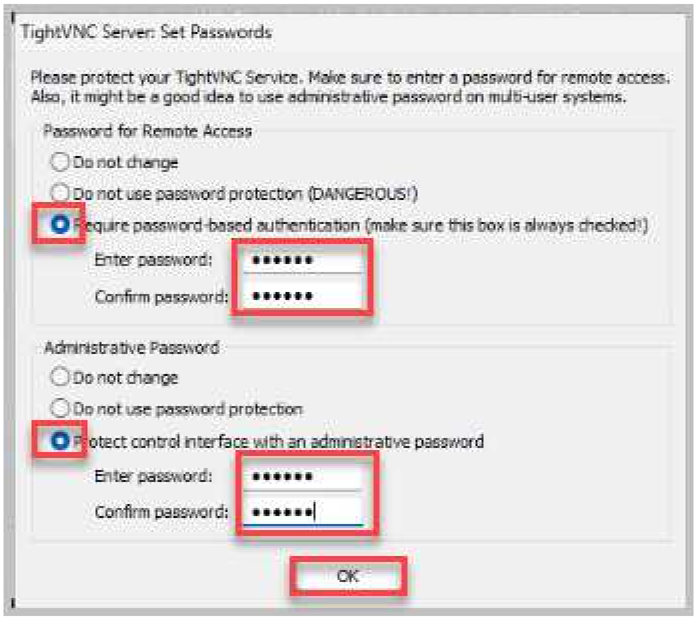
10. Click the **Install** button.



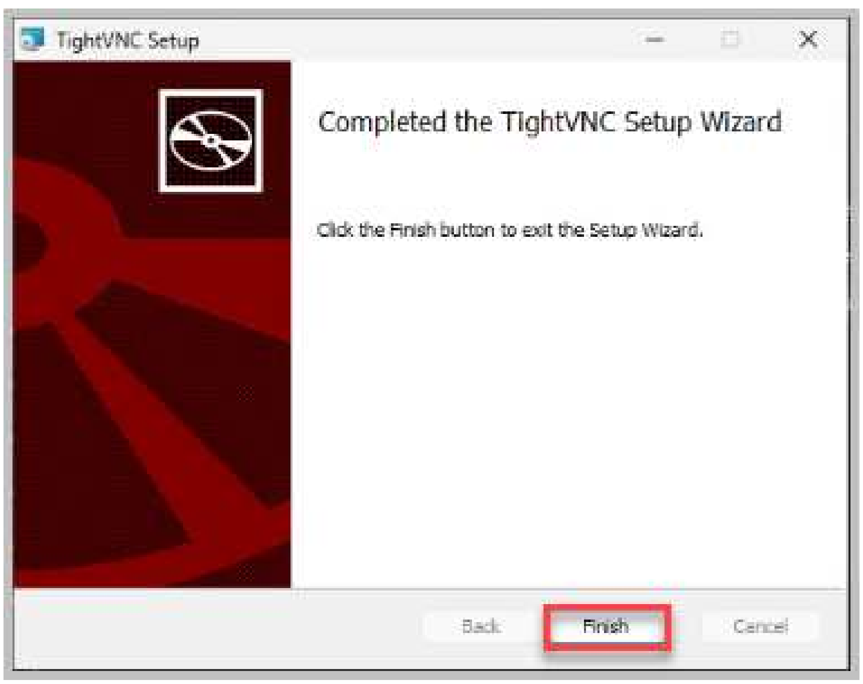
11. When prompted for user account control, click **Yes** to continue.



12. In the *TightVNC Server: Set Passwords* window, select the radio button for **Require password-based authentication (make sure this box is always checked)** and type VNCpwd for both password fields. Next, select the radio button for **Protect control interface with an administrative password** and type VNCpwd for both password fields. Click **OK**.

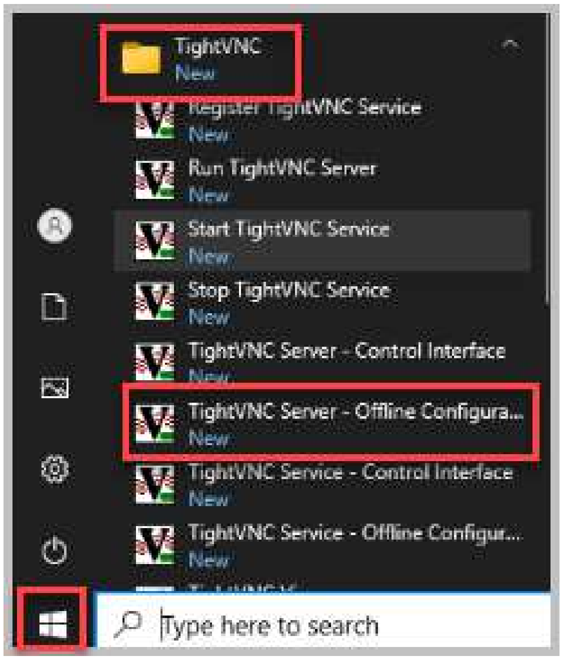


13. Once finished, click the **Finish** button.

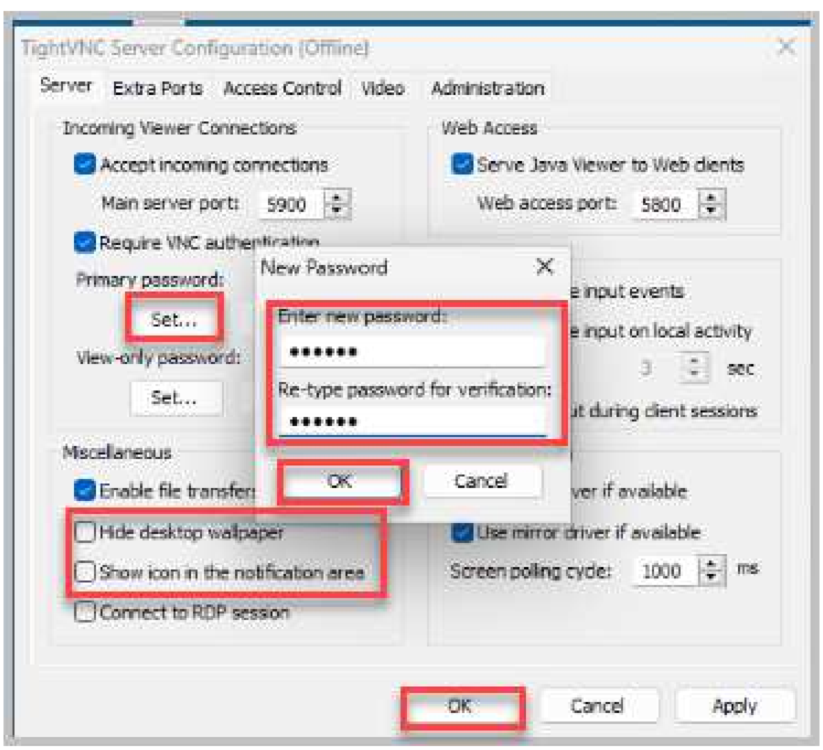


14. Close the *Toolbox Folder.*

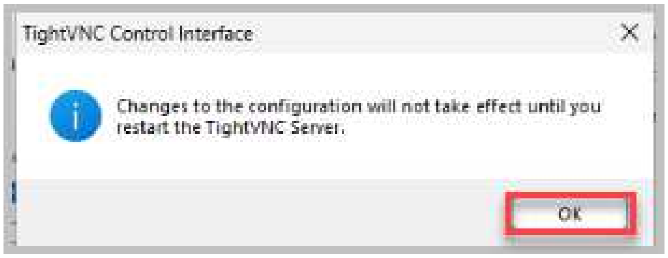
15. Now the *VNC* server is installed, and by default, the *VNC* server service has already started and is configured to start automatically when the machine boots. Click the **Start** button and navigate to **All Apps > TightVNC** and click on **TightVNC Server- Offline Configuration**.



16. On the *Server* tab, in the *Miscellaneous* pane, ensure that the checkboxes for **Hide desktop wallpaper** and **Show icon in the notification area** are unchecked. In the *Incoming Viewer Connections,* under *Require VNC authentication,* select **Primary password. Type** `VNCpwd` **for both new and retyped passwords** for verification. Click **OK** to close *the New Password.* Click **OK** to close *TightVNC Server Configuration (Offline).*



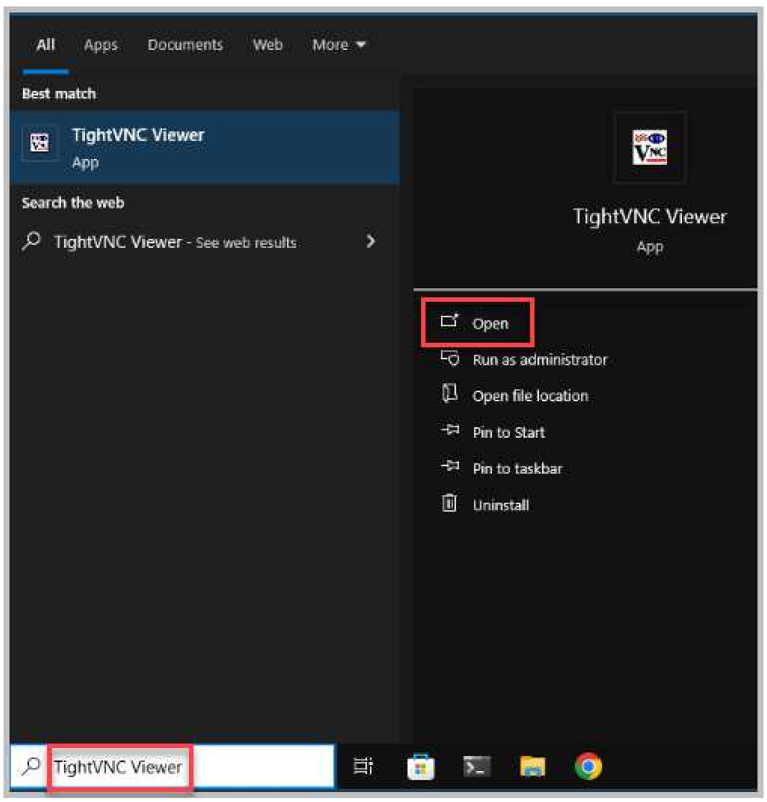
17. Select **OK** to accept *Changes to the configuration will not take effect until you restart the TightVNC Server.*



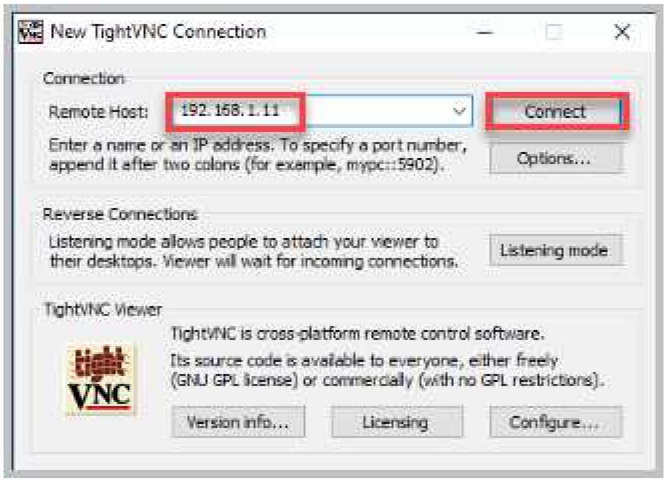
18. Leave the *Win10-OS* window open to continue with the next task.

2.3 Using TightVNC Viewer on Win10-OS

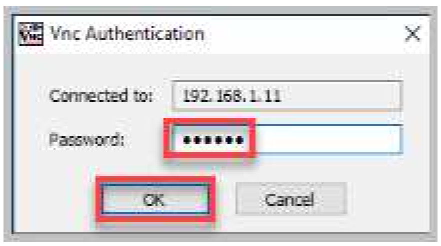
1. Click the **Windows Search** icon and type `TightVNC Viewer` in the search field. Click **Open.**



2. In the *New TightVNC Connection*, type `192.168.1.11` in the *Remote Host* field. Click **Connect**.



3. When prompted for authentication, type `VNCpwd` in the *Password* field. Click **OK**.



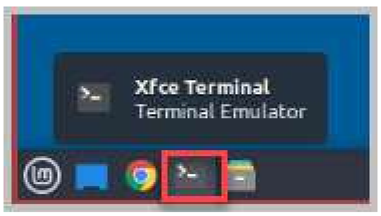
4. Notice that a successful remote session is made with the *Win11-OS* system.
5. Close *the TightVNC viewer by selecting X in the top right corner of the Win11-OS TightVNC viewer.*

3 Setup and Connect to Linux with TightVNC

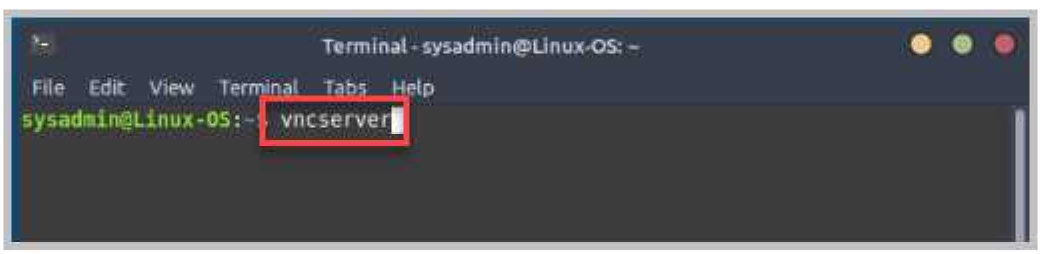
3.1 Setup TightVNC Server on Linux-OS

1. Connect to the **Linux-OS** console.

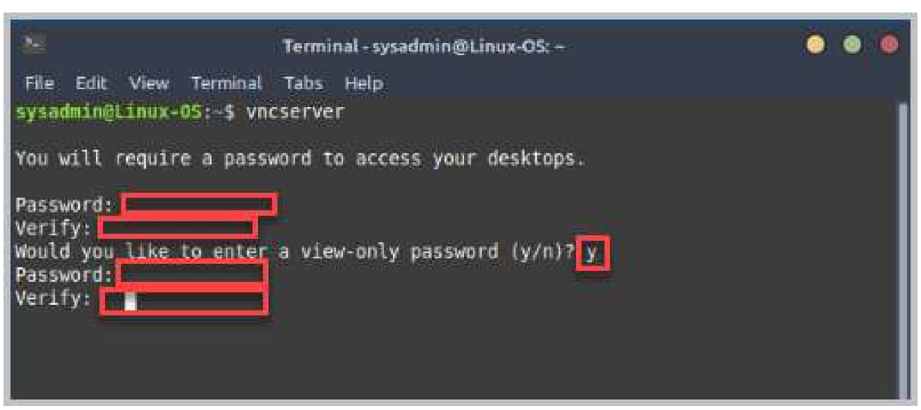
2. Select the **Xfce Terminal** on the taskbar.



3. Type **vncserver** and hit **enter** on your keyboard.



4. Type **VNCpwd** for the password to access the *Linux-OS Desktop.* Retype the same `VNCpwd` for verify password. Select **y** to enter a view-only password, use `VNCpwd` for both the password and verify password.



5. If not open, reopen the **Xfce Terminal.**

6. Type `ps aux | grep vnc` and select **Enter.** You will see that 1 vncserver is running.

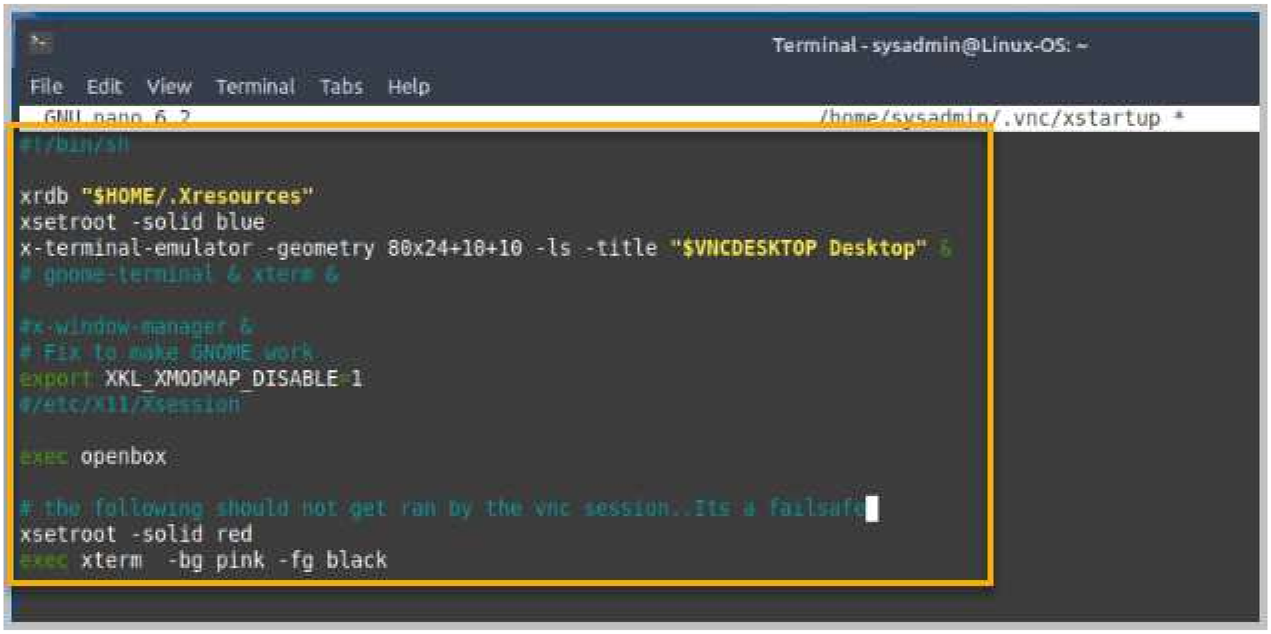
7. Type `vncserver -kill :1` to stop the vncserver, select **Enter.**
8.

9. Verify that *vncserver is not running* by typing `ps aux | grep vnc`.

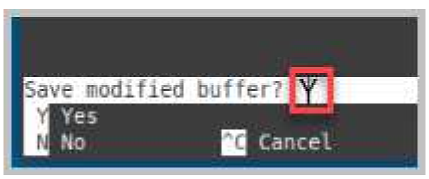
10. Type `nano /home/sysadmin/.vnc/xstartup` and select **Enter.**

11. In the *xstartup file,* you will delete everything existing in the *xstartup file* and enter the following items as listed below:

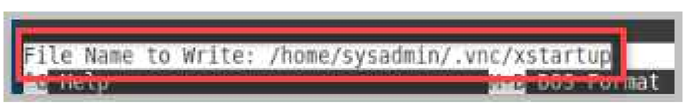
```
#!/bin/sh

xrdb "$HOME/.Xresources"
xsetroot -solid blue
x-terminal-emulator -geometry 80x24+10+10 -ls -title “$VNCDESKTOP DESKTOP”
&
# gnome-terminal & xterm &

#x-window-manager &
# Fix to make GNOME work
export XKL_XMODMAP_DISABLE=1
# /etc/X11/Xsession

exec openbox
# the following should not get ran by the vnc session.. It's a failsafe
xsetroot -solid red
exec xterm  -bg pink -fg black
```

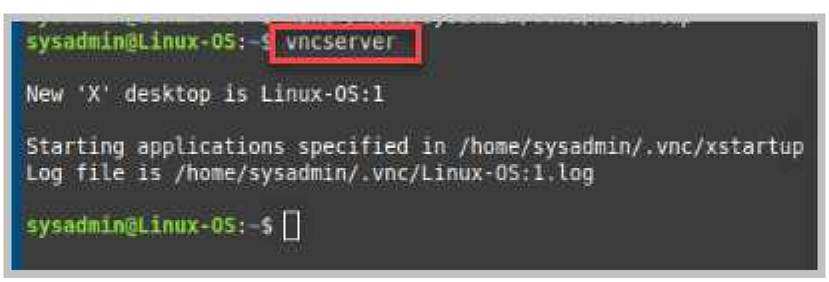
12. After verifying the information is entered correctly, on your keyboard, select **Ctrl + X.** This stops the editing of the *xstartup file.* Select **Y** to save the file.



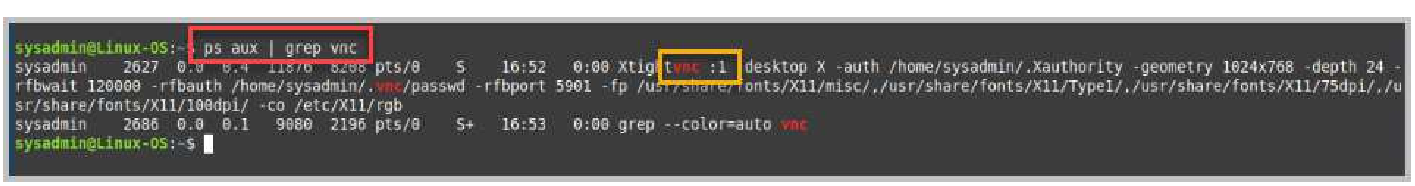
13. Hit **Enter** on your keyboard to save the filename to the default location of *home/sysadmin/.vnc/startup.*



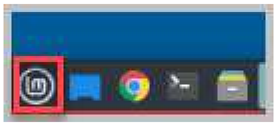
14. In the *Terminal Window,* type **vncserver.**



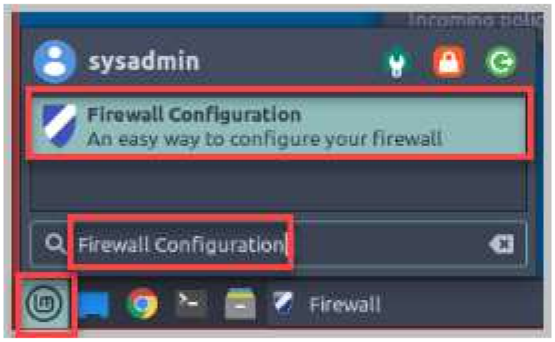
15. To verify *vncserver* is running, type **ps aux | grep vnc.** Close the terminal window.



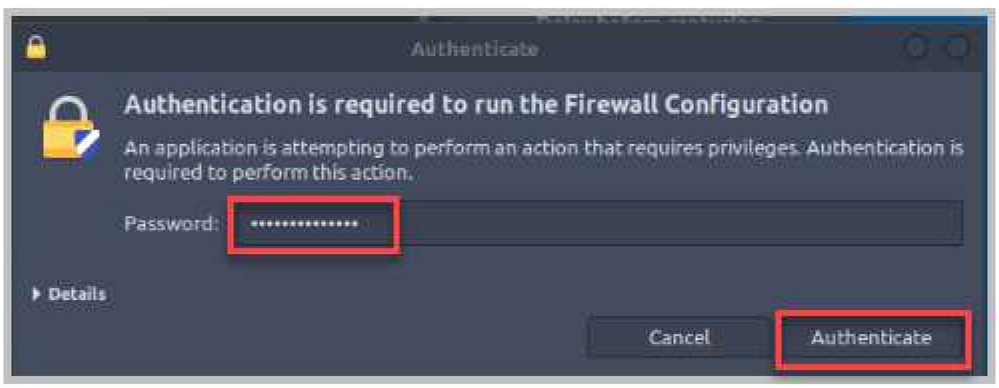
16. Select **Start** on the taskbar.



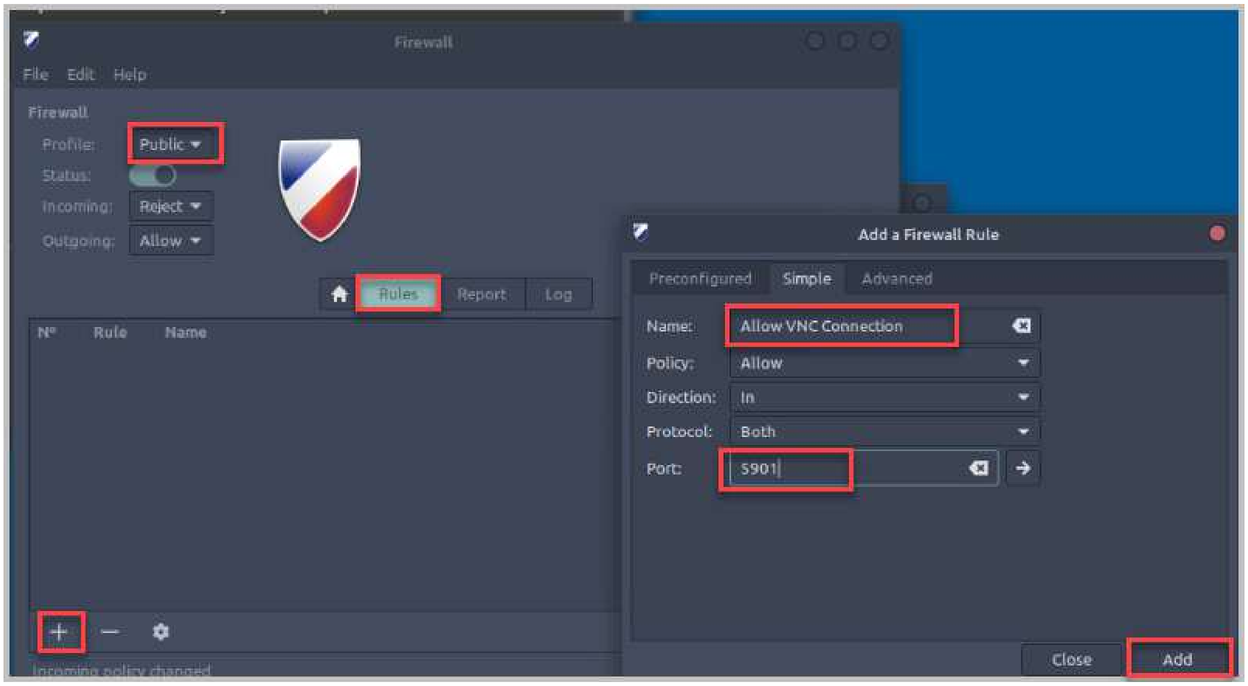
17. Type **Firewall Configuration,** select the *Firewall Configuration App.*



18. Type the password `NDGlabpass123!` and select **Authenticate.**

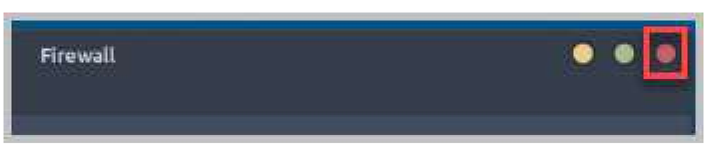


19. Select the **Rules** tab. Select **Public** for the *Profile,* select **+** to add a Firewall Rule. Select the **Simple** tab, type `Allow VNC Connection` for the Name, and type **5901** for the port to allow. Select **Add.**



20. After selecting *Add, two rules* have been created. Select **Close** for the *Add a Firewall Rule.*

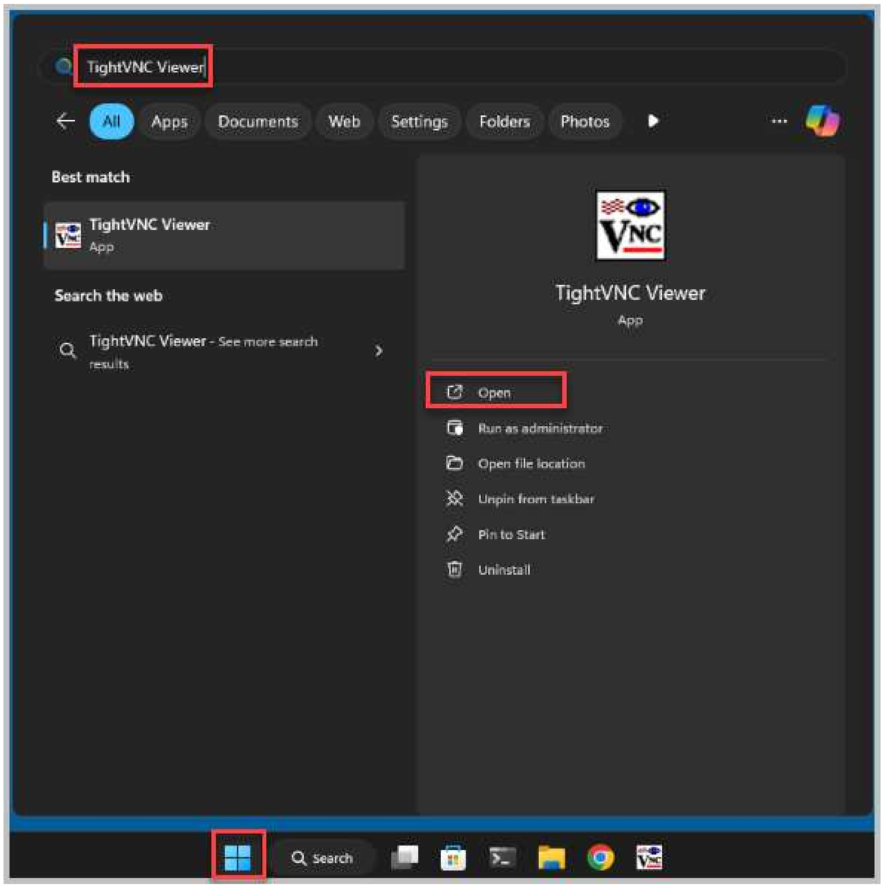
21. Select the **X** in the top right of the *Firewall* window.



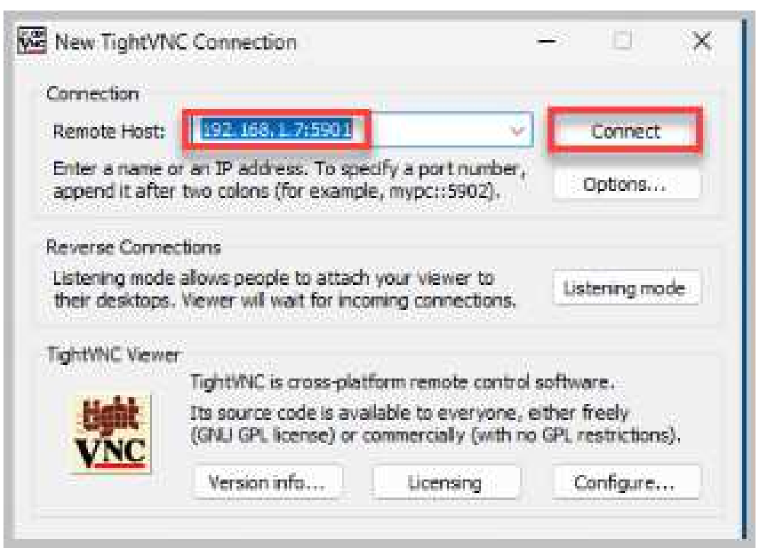
3.2 Connect to Linux-OS from Win11-OS

1. Open a console connection to the **Win11-OS** system.

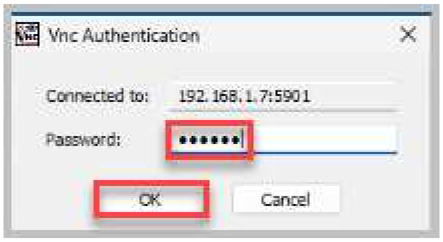
2. Select **Start** on the taskbar and type **TightVNC Viewer.** Select **Open.**



3. Type `192.168.1.7:5901` for the remote host address. Select **Connect.**



4. Type **VNCpwd** for the VNC Authentication Password and select **OK.**



5. You have successfully connected to the *Linux-OS* system from *Win11-OS* utilizing the *TightVNC Viewer.*
6. The lab is now complete; you may end the reservation.

